

# RAYNEX

## ONE SUN. FIVE JOBS. PAID TO SAVE WATER.

A farmer-owned solar hub for 20 smallholders on 8 ha, piloted across three farm zones of Nashik. One Nashik. One Schneider-run array keeps the cold room safe, waters by the soil, freezes ice, dries onion dries onion and sells the rest; drones and robots join as a shared service. The DISCOM shares the shares the power cost it saves when the water stays in the ground.

💧 **1,960 m<sup>3</sup>**

**Water Left In The Ground**  
per farmer a year, -35%

🍅 **752 kg**

**Harvest Saved From Rot**  
per farmer a year, loss cut by a third third

📈 **₹32,460**

**Kept Every Year**  
per farmer while the loan runs,  
₹55,077 after

🕒 **Payback 2.5 Years**

☀️ **72% Sun Used On Site**

✅ **₹0 Farmer Capital**

📍 **Pilot: Nashik, 3 Zones**

### FIVE JOBS, ONE ARRAY, IN PRIORITY ORDER

❄️ **1 Cold Room**

💧 **2 Drip Irrigation**

📦 **3 Ice Store**

☀️ **4 Dryer**

🔌 **5 Grid Export**

\$ **stack** ATV320 Solar · Modicon M262 · LoRaWAN · Sarvam voice, 22 languages · drone + robot hire

## MEET THE TEAM



Abhinav Suresh



Arpan Das



Prince Panja



Shubham Mali

## IIM INDORE

Indian Institute of Management Indore

## The Four Personas

**Ramesh wants sleep, Sunita wants safety, Pooja wants a job, Anil wants a smaller bill. One hub has to serve all four.**

## 📌 E1: Problem Statement · Personas



## Ramesh Shinde, 41

Farmer, Niphad, Nashik  
0.4 ha · black soil · onion, tomato

## Today

- ☾ Power is free, so he floods at **2 am** when the the feeder comes
- 💧 Pumps **5,600 m<sup>3</sup>** a year in a critical block, **97% 97%** extracted
- 🍌 Loses about **1.9 t**; sells at Lasalgaon on harvest day

*"Water my field without a 2 am trip and sell on a sell on a good day."*

**Wants: sleep, +₹32,460 a year**



**Sunita Gaikwad, 36**

Woman farmer, Yeola, Nashik  
shallow soil · onion · SHG member

## Today

- ⚠️ Waters alone at night while her husband migrates for work
- 🗣️ Speaks Ahirani, cannot read an app or an SMS
- 🏠 Onion rots in a shed; she sells to the first trader

*"Water by day, store my onion, tell me in my  
my own tongue."*

**Wants: safety, voice, storage**



**Pooja Pawar, 22**

ITI electrician, Igatpuri  
smartphone · Marathi, English

## Today

- 🚚 Only local option: a factory shift on the highway
- 🕒 Nearby solar pumps break and sit idle for weeks
- 📱 Wants live alerts, not a call from an angry farmer farmer

*"A skilled job near home that pays every month."  
month."*

**Wants: ₹15,000 a month, a career**



**Er. Anil Deshmukh, 38**

Assistant Engineer, MSEDCL  
Niphad sub-division · feeders

## Today

- 📄 The state pays ₹14,761 cr a year for free farm power
- ⚡ Farm load peaks at night on weak rural feeders
- 👁️ No proof of how much water that power buys

*"Cut the farm power bill without a farmer revolt."*

**Wants: a smaller bill, proof**

## Contrasting Needs, One Hub: What Each Feature Gives Each User

Exhibit F6, Slide 11

| Feature                  | Ramesh, Farmer            | Sunita, Woman Farmer                | Pooja, Technician            | Anil, DISCOM                    |
|--------------------------|---------------------------|-------------------------------------|------------------------------|---------------------------------|
| Daytime Solar Drip       | No 2 am trips; –35% water | Waters by day, near home            | Fewer night fault calls      | Night load moves to day         |
| Cold Room, Sell-Or-Store | Holds tomato for + ₹3/kg  | Onion stored, not dumped            | Watches room temperature     | Cold load runs on own solar     |
| Voice In 22 Languages    | Marathi call at 7 am      | Ahirani voice, no reading           | Marathi and English app      | English portal                  |
| Bachat Bonus             | Share of ₹46,968 via FPO  | Same share, own vote                | Meters she keeps prove it    | Pays only on metered saving     |
| Plot Score And Alerts    | One spoken action a day   | Drone Didi sprays weak patches only | Alert to her phone in 15 min | Feeder data without site visits |
| FPO Owns The Hub         | ₹0 capital                | One member, one vote                | Paid ₹15,000 a month         | One counterparty, 20 farmers    |

## Design Rules From Four Users



### Voice Before Screens

Ahirani and Marathi first



**Daylight Before Night**  
No pump runs after dark



**Proof Before Payment**  
- Meters, not trust



### Owners Before Automation

Recommend mode, one vote

Water, sun and harvest all leak for the same reason: the next unit costs the farmer nothing and saves him nothing.

Free marginal water, idle solar and no farm-gate cold cost India ₹2.8 lakh crore a year.

Leak 1: Water Nobody Pays For

Groundwater extracted, % of recharge, Nashik Nashik

Sinnar99.9%

Niphad97.4%

Deola95.0%

Baglan83.6%

Yeola79.0%

District58.5%

📍 Maharashtra pays ₹14,761 cr a year so 44.06 lakh pumps run free

₹ Next hour of pumping costs the farmer ₹0; the DISCOM pays ₹5.54 a unit

🏠 States spent ₹1.89 lakh cr on power subsidy in FY25 (CAG)

📉 India extracts 245.6 BCM a year; 11% of blocks over-exploited

→ Exhibit F2 Subsidy Graph, Slide 11

Leak 2: Solar That Sleeps

Standalone solar pump, days a year

150 pumping215 idle

35%

40%

Solar used on site, Dhundi pilot

🔦 Solar fixes the fuel bill but makes the next hour of pumping cheaper still

📍 SKY paid for surplus power; selling water paid ₹6 to ₹8 a unit, so 60% kept pumping

⚡ No battery the farmer can afford pump-sized lithium wrecks payback

🕒 215 idle days: the asset earns nothing for 59% of the year

→ Exhibit F4 Lessons Abroad, Slide 11

Leak 3: Harvest That Rots

Loss from field to market, % of output

Fruit, max15.1%

Tomato11.6%

Onion7.3%

Cereals5.9%

🍅 Tomato is the worst vegetable; loss across crops ₹1.53 lakh crore a year

📖 Nashik grew 58.8 lakh t of onion in 2024-25; Maharashtra is 35.5% of India

🚚 No farm-gate cold, so Ramesh sells at Lasalgaon on harvest day

What India loses a year, ₹ crore

Power subsidy1,30,000

Harvest loss1,53,000

→ Exhibit A5 Loss Baseline, Slide 12

First Principles: Six Questions

Why over-pump?  
Power is free up to 7.5 HP, so the next hour costs him nothing

Who loses most?  
The state, ₹1.3 lakh crore a year, so the state should pay for water saved

Why do tomatoes rot?  
No farm-gate cold; the cheapest battery is a tank and ice

Why no adoption?  
Cost, trust and reach: share the asset, speak the advice

Why is efficiency not enough?  
Saved water gets spent on more land; cap the total, not water per ha

Who should own it?  
The farmers, one member one vote: Denmark had 1,163 co-op dairies by 1909

Root cause: free marginal water, and saving it pays no one.

Climate Makes It Worse, Rails Make It Fixable

☁ Monsoon 14 to 15% short in 2026, with more heat

🌐 Farms: 40%+ of jobs, ~90% of freshwater

🏭 Schneider: 15 to 20% of produce value lost

📖 Nashik alone spans 550 to 3,400 mm of rain

Digital rails already built, crore

Crop plots surveyed31.3

Farmer IDs10.31

SMS monsoon alerts3.88 (31 to 52% acted)

Kisan e-Mitra queries0.93 in 11 languages

🌾

🖥

One farmer-owned array, a Schneider router that sends every unit of sun to the job worth most, and a bonus for water saved.

Nine parts, one FPO owner, one DISCOM contract: every Schneider goal gets a number.

E2: Solution · E3: Alignment

The Hub, In Priority OrderExhibit A3, Slide 12



 1 Cold Room

5 t room guards up to 250 crates; 2,700 kWh

 2 Irrigation

Soil-gated drip, 8 zones; 15,430 kWh kWh

 3 Ice Store

Midday ice keeps the room cold at night

 4 Dryer

Onion and chilli; 300 kWh

 5 Grid Export

7,070 kWh at ₹3.05, plus the bonus

72% of the sun used on site, against 35% for a solar pump alone

Nine Components, One Contract

 C1

Energy Router

15 kWp, ATV320 drive, picks the load

 C2

Raised Array

3.5 m stilts over the nursery, ≤30% shade

 C3

Soil-Gated Drip

–35% water, +20% yield, 50 m³ tank

 C4

Bachat Bonus

DISCOM pays 50% of the subsidy saved

 C5

Cold-Chain Ladder

₹6,000 chambers, then a 5-tonne room

 C6

Plot Score

One 0 to 100 score per plot, daily

 C7

Drone Didi

SHG pilots spray only flagged patches

 C8

Voice First

Voice in 22 languages, from Marathi

 C9

FPO Operator

20 farmers, 8 ha, one member one vote

New: C1 router chooses the load itself, and C4 pays for water not pumped.

→ Exhibit F5 Blue Ocean, Slide 11

Schneider's Four Goals, Each With A Number

1

Cut Energy And Water Intensity

–35% water; 3.57 → 6.59 kg/m³; 100% solar

2

Minimise Post-Harvest Loss

Tomato loss 11.6% → 7.7%; 15.04 t saved per hub

3

Empower The Smallholder

₹0 capital, voice in Marathi, +₹32,460 a year

4

Strengthen Climate Resilience

Tank, ice, AquaCrop; 39,200 m³ a hub kept underground

Schneider's Five Starter Ideas: All Five In

 Solar precision irrigation C1, C3

 AI crop health C6

 Solar cold chain + logistics C5

 Vernacular voice C8

 Aggregation, co-ops C9

Schneider's Condition, Answered

 Software: Energy router + Plot Score

₹ Business model: FPO-owned hub, Bachat Bonus, AIF loan

 Farmer tools: Missed call, voice, SMS, a village operator

Why Not A Simpler Fix?

| Option                | Sun Used On Site   | Water                            | Harvest             | Who Pays                    |
|-----------------------|--------------------|----------------------------------|---------------------|-----------------------------|
| Solar pump alone      | 35%, idle 215 days | Pumps more: 60% unchanged in SKY | No cooling          | Farmer + subsidy            |
| Free grid pump, today | 0%                 | Floods at night, free            | No cooling          | State: ₹14,761 cr a year    |
| Cold room alone       | Grid power         | No change                        | Loss cut            | Farmer                      |
| RAYNEX hub            | 72%, five loads    | –35%, capped by budget           | Loss cut by a third | FPO, AIF loan, DISCOM bonus |

Three design artifacts: a service blueprint for the season, the missed-call voice flow, and four offline app screens.

No smartphone needed: one missed call, one answer in their own language, one action a day.

E4: User Journey · D3: Design Artifacts

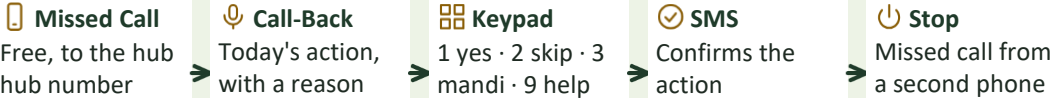
Service Blueprint: Five Lanes Through One Season

Exhibit F6 JTBD And Rogers, Slide 11

| Lane         | Before Sowing                                        | Growing Season                                             | Harvest                                           | Off-Season                                           |
|--------------|------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------|------------------------------------------------------|
| Farmer       | Joins the FPO with his Farmer ID; voice onboarding   | Morning voice call; approves or overrides (recommend mode) | Drops crates at the hub; hears sell-or-store      | Input-cost alerts; books seed and fertiliser in bulk |
| Operator     | 90-second soil scan; installs probes at 15 and 45 cm | Clears alerts within 24 h; Pooja backs 5 hubs              | Weighs every lot in and out on the platform scale | Cleans panels; services the cold room                |
| Router, Edge | Loads the crop plan and the season water budget      | Waters to the soil balance; caps pumping at budget         | Cold room first, then ice store                   | Dryer runs, surplus exported                         |
| Cloud, Data  | AquaCrop plan; NDVI zones from Sentinel-2            | Plot Score 0 to 100, daily, per plot                       | Shelf life per lot; route and truck pooling       | Season report and metered bonus proof                |
| Partners     | AIF bank loan; KVK Nashik                            | Drone Didi spray bookings                                  | Lasalgaon buyers, eNAM                            | DISCOM pays the bonus; insurer gets consented data   |

Missed-Call Voice Flow: Marathi, Ahirani, Hindi At Launch

"Rameshram, aaj zone 2 la 40 minit paani dya. Udaya paavsaachi shakyaata 70 takke." takke." (Marathi: water zone 2 for 40 minutes; 70% chance of rain.)



Four App Screens: Flutter, Offline-First, Icon-Led (illustrative)

1 Today

Z1 Z2 Z3 Z4

Pump: solar on

Rain tomorrow 70%

Do: zone 2, 40 min

Press 1: Approve

2 Store Or Sell

Lot 14: 4 days left

Today ₹10/kg

Hold: +₹3/kg

Do: store, sell Friday

Store This Lot

3 Earnings, Budget

Sun: made, used, sold

Water vs budget

Bonus: on track

See This Season

4 Crop Health

Plot Score 78

NDVI: 2 weak patches

Leaf photo check

Do: book Drone Didi

Book Drone Didi

Ramesh's Day, Then  
And Now  
Zero-night trips

2 AM

Wakes for feeder power,  
floods the field

Asleep: tank and ice did the  
night shift

0 Night Trips

7 AM

Guesses what to water

Voice call: zone 2, 40  
minutes

1 Action A Day

11 AM

Sun on an idle pump

Router runs drip on solar, by  
the soil

−35% Water

2 PM

Picks tomatoes in the heat

Crates into the cold room

250 Crates Safe

5 PM

Sells at today's price

Sell-or-store says hold

+₹3/Kg On Stored Lots

9 PM

Tomatoes start to soften

Room on ice; ledger updated

Loss Cut By A Third

A Schneider ATV320 drive and a Modicon M262 router run every load; data decides on the farm; money loops via FPO, bank, MSEDCL.

Energy goes where it is worth most, data decides on the farm, and money pays for water left in the ground.

Where The Sun Goes, kWh A Year

25,500 kWh

Irrigation

15,430 · 60%

Cold Room

2,700 · 11%

Dryer

300 · 1%

Grid Export

7,070 · 28%

72% used on site, against about 35% for a solar pump at Dhundi

Fail-Safe By Design

✓ Edge controller runs pump, cold band and budget cap **on the farm**

⚡ A network drop **never** stops water or cooling

🔌 Manual override on pump and cold room

👤 **Pooja**, Hub Mitra for 5 hubs, fixes alerts within 24 h

System Architecture: Five Layers, Three Flows

Energy

Data

Money

Exhibit A3, Slide 12

L5  
People

Farmers: Voice, 22 Languages

1 Action A Day

L4  
Intelligence

FAO-56, AquaCrop, NDVI

Plot Score 0 To 100

Daily, Per Plot

L3  
Post-Harvest

S4S Dryer, Chambers

300 kWh Dryer

Ecofrost 5 t, Thermal

2,700 kWh Cold

250 BLE Crate Tags

4G Sync

L2  
Field

Drip, 8 Zones, Fertigation

-35% Water

10 HP AC Pump, 50 m³

15,430 kWh

LoRa Probes, Flow, Piezo

Every 15 Min

UG65 Gateway + Edge

L1  
Energy

15 kWp TOPCon DCR, Stilts

25,500 kWh/yr

ATV320 Solar, 7.5 kW

Modicon M262 Router

Priority 1 To 5

EM6400NG + Net Meter

7,070 kWh Out

FPO: Owns Hub, 20 Farmers

AIF Bank: ₹4.52 L A Year

₹46,968 Bonus

MSEDCL Portal, Insurer

Proof

AWS Mumbai, 4G Sync

Built On Schneider Parts

ATV320 Solar: MPPT, Dry-Run Guard, To 50°C

SMS On/Off + EcoStruxure Livelihood

Modicon M262 Runs The Router

EM6400NG Meters: Bonus Proof

Up To 30% Less Pump Energy

SCHNEIDER INSIDE

Router Priority, Every Few Minutes

✳️ 1 Cold Room: ₹53 of power guards ₹50,000 of crop

💧 2 Irrigation: zones flagged dry

📅 3 Ice Store: midday ice for night

☀️ 4 Dryer: onion and chilli

🔌 5 Grid Export: at ₹3.05 a kWh

Money Loop, Per Hub A Year

₹ Farmer Fee ₹30,117 → FPO

🏠 FPO → Bank ₹4.52 L AIF

📄 MSEDCL → FPO ₹46,968 Bonus

🏛️ MSEDCL → FPO ₹21,564 Export

🏠 Subsidies ₹9.10 L at 80%

Ramesh's Ledger every year

👤 **Ramesh, 41**  
0.4 ha in Niphad, Nashik

💧 **1,960 m³**  
water left in the ground

🍅 **752 kg**  
tomato and onion saved

₹ **₹32,460**  
kept a year during the loan

🌙 **0**  
night trips to the pump

Off-the-shelf hardware, open-source software, and a voice stack that speaks 22 Indian languages by Tier 3.

🕒 D4: Prototype · E5: Technical Approach

| Hardware Bill Of Materials                                                                                 |                              |              | Exhibit A7, Slide 12 |
|------------------------------------------------------------------------------------------------------------|------------------------------|--------------|----------------------|
| Component                                                                                                  | Model And Spec               | ₹ Indicative |                      |
| PV, 15 kWp                                                                                                 | 26 × 580 Wp TOPCon, DCR      | 3.6 to 4.1 L |                      |
| Stilts, 3.5 m                                                                                              | HDG steel, 80 µm coat        | 2.1 to 2.7 L |                      |
| Pump                                                                                                       | 10 HP AC submersible         | 3.5 to 4.5 L |                      |
| Solar Drive                                                                                                | Schneider ATV320 Solar       | 50 to 75k    |                      |
| Router PLC                                                                                                 | Schneider Modicon M262       | 25 to 60k    |                      |
| Meters × 3                                                                                                 | Schneider EM6400NG           | 7 to 13k ea  |                      |
| Soil Nodes × 8                                                                                             | LoRaWAN moisture + EC        | 8 to 15k ea  |                      |
| Flow Meter                                                                                                 | Kritsnam Dhaara, ultrasonic  | 31 to 38k    |                      |
| Borewell Level                                                                                             | Piezometer, telemetry        | 18.5k        |                      |
| Gateway, Edge                                                                                              | Milesight UG65 + 4G router   | 60 to 85k    |                      |
| Weather Station                                                                                            | Rain, heat, RH, wind, sun    | 0.5 to 1.5 L |                      |
| Cold Room, 5 t                                                                                             | Ecozen Ecofrost, 30 h backup | 11.4 to 15 L |                      |
| Dryer                                                                                                      | S4S conduction, 100 kg       | 3.6 L        |                      |
| Drip, 8 ha                                                                                                 | Jain or Netafim, fertigation | 8.96 L       |                      |
| Gross ₹37.16 L a hub; gold rows are Schneider parts. Tenders now buy a solar pump at pump at about ₹2.4 L. |                              |              |                      |

| Software Stack, Top To Bottom                                                        |                                                                                                                      |
|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| APPS                                                                                 | Flutter app, offline, under 42 MB · IVR · WhatsApp<br>WhatsApp · FPO dashboard · MSEDCL portal portal                |
| AI, VOICE                                                                            | Sarvam speech in and out · Sarvam-105B on ICAR advice · Plot Score model                                             |
| BACKEND                                                                              | Python FastAPI · TimescaleDB · AquaCrop-OSPy + OSPy + FAO-56 · Sentinel-2, IMD, Agmarknet, Agmarknet, AgriStack APIs |
| CLOUD, IoT                                                                           | AWS Mumbai, MeitY-empanelled · AWS IoT Core, \$1 per million messages · ChirpStack LoRaWAN · ThingsBoard             |
| EDGE                                                                                 | Python and Node-RED on the gateway · MQTT store-and-forward · pump, cold band and budget cap run locally             |
| DEVICE                                                                               | LoRa soil nodes · Modbus meters · ATV320 · M262 · BLE crate tags                                                     |
| 🔒 DPDP Act 2023: farmer-owned data, consent before any any insurer or lender sees it |                                                                                                                      |
| 📶 A network drop never stops water or cooling                                        |                                                                                                                      |

| Voice In 22 Languages                                                                                    |                                                                                            | Exhibit A10, Slide 12                                                                                               |
|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Speech In</b><br><b>Sarvam Saaras v3</b><br><br>22 languages, phone<br>phone audio, ₹30 an<br>an hour | <b>Reasoning</b><br><b>Sarvam-105B</b><br>22 languages,<br>Apache 2.0,<br>grounded on ICAR | <b>Speech Out</b><br><b>Bulbul v3, Parler-<br/>TTS</b><br>11 + 21 languages,<br>languages, ₹30 per<br>per 10K chars |
| <b>Tier 1, Launch</b>                                                                                    | Marathi, Hindi, Ahirani (fine-tuned on <b>5 to 7 h</b> of speech)                          |                                                                                                                     |
| <b>Tier 2, Month 4</b>                                                                                   | Telugu, Kannada, Gujarati, Bengali, Tamil, Odia,                                           |                                                                                                                     |
| <b>Month 4 To 9</b>                                                                                      | Punjabi, Malayalam; Bhili, Bundeli, Bhojpuri, Malvani                                      |                                                                                                                     |
| <b>Tier 3, All 22</b>                                                                                    | IndicConformer + IndicTrans2 + Parler-TTS; TTS; Bhashini as failover                       |                                                                                                                     |
| ₹                                                                                                        | About <b>₹7 to 8</b> per 3-minute call; cached answers cut the costliest part, speech out  |                                                                                                                     |
| 👥                                                                                                        | Demand is proven: Kisan e-Mitra answers <b>8,000+</b> queries a day in <b>11</b> languages |                                                                                                                     |
| 📞                                                                                                        | Low speech confidence? Keypad menu, then an agronomist calls back                          |                                                                                                                     |
| ⚠️                                                                                                       | Bharat-VISTAAR speaks only Hindi and English so far; BharatGen Param2 is non-commercial    |                                                                                                                     |

D4 Software  
Prototype By 22 Nov  
Hand-coded, no hardware

⚡ 1 Router Simulation  
Hourly Nashik sun for a year; 72% vs 35% on site  
  
out: Monthly kWh by load

💧 2 Irrigation Scheduler  
FAO-56 + probes + rain-skip; AquaCrop-OSPy runs  
  
out: Minutes per zone, daily

🗣️ 3 Voice Demo  
Marathi and Ahirani missed-call IVR on a test number  
  
out: One spoken action + SMS

🎯 4 Plot Score Demo  
Sentinel-2 NDVI for one Niphad village + mock probes  
  
out: Score and weak patches

🏠 5 Sell-Or-Store  
Onion and tomato storage + Lasalgaon prices  
  
out: Hold or sell, best market

No smallholder can own a drone or a robot; an FPO hub can, and rents them out by the acre, charged on the same solar array.

China flies 251,000 farm drones; India's Drone Didi scheme has delivered 1,094. A shared hub closes that gap.

E6: Innovation · Automation Layer

Drones: Live From Year 1

Exhibit A9, Slide 12



| Metric         | Manual Knapsack | Drone, 10 to 16 L |
|----------------|-----------------|-------------------|
| Cost per acre  | ₹800 to 1,500   | ₹400 to 800       |
| Water per acre | 200 to 250 L    | 10 to 15 L, -94%  |
| Speed          | a day per acre  | 1 acre in 15 min  |

- ☉ Sprays only the weak patches the **Plot Score** flags; scouts with a with a multispectral camera
- ⚡ Batteries charge on hub solar at midday, when the router has surplus
- ₹ **Funding:** Namo Drone Didi **80%** up to ₹8 L; SMAM **75%** for FPOs FPOs
- 👤 Pilot earns ₹400 to 600 an acre; Igatpuri hub also hauls produce produce downhill

Robots: Phased In From Year 2

**Y2 Spot-Spray Robot**  
Camera finds each weed or sick plant  
Niqa: **1.48 lakh** acres in 2024-25, Akola, MH; MH; inputs -60%

**Y3 Grading Line, Robot Arm**  
Sorts onion by size and rot into crates  
Feeds shelf life per lot; cuts packing labour

**Y5 Autonomous Weeding**  
US See & Spray: **5 mn+** acres, ~50% less herbicide  
Hired, never owned: the FPO runs a custom-hiring centre

- 🔄 Plot Score flags a patch → app books a machine → it charges on hub solar → works → proof logged
- 🏠 AIF funds custom-hiring centres and drones; machines are AIF-eligible assets

India Vs The World

Exhibit F4, Slide 11

**Farm mechanisation, %**

|        |     |
|--------|-----|
| USA    | 95% |
| Brazil | 75% |
| China  | 60% |
| India  | 47% |

**Farm drones in use, log scale**

|       |                  |
|-------|------------------|
| China | 2,51,000         |
| India | 1,094 Drone Didi |

**Netherlands** Greenhouse tomato **500+ t/ha** vs India ~25

**Israel** **90%** micro-irrigated vs India **5.5%**; flying fruit pickers

**Japan** Drone-sprayed area up **175x** in 5 years; farmers average **67.8**

**USA** FAA lets **3-drone** swarms spray **150** acres an hour

⚠️ India: **46%** of workers, **16%** of GDP; Drone Didi at **7%** of its **15,000** target

Automation Roadmap

Machines rented; land stays the farmer's

Y1 🛩️ 1 Drone, 1 Pilot

Spray and scout; Sunita's SHG flies it

₹8 L subsidy cap

Y2 ☉ Spot-Spray Robot

Hired per acre from a cluster of hubs

inputs -60%

Y3 🏠 Grading Line

Onion and tomato sorted at the cold room

shelf life per lot

Y4 🚚 Autonomous Cart

Moves crates field to cold room

labour -30 to 40%

Y5 ☀️ Weeding Robot

Rented across 25+ hubs

herbicide ~-50%

# Pay To Save Water, And What It Returns

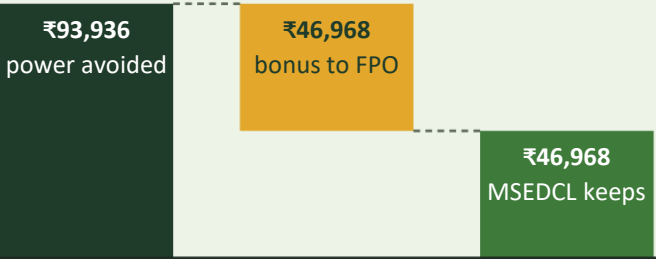
Baseline: a Niphad smallholder, tomato in kharif and onion in rabi, flood irrigation on a free-power grid pump, no farm-gate cooling.

A metered bonus makes saving the best move; a hectare returns ₹1,37,693 a year on 35% less water.

✓ E6 Innovation · D5 Benefit · E7 Impact

### What Is New: The Bachat Bonus

Per hub: 16,956 kWh × ₹5.54 power purchase cost



- 1 Borewell sensor + flow meter set the season's m³ budget budget
- 2 Controller caps pumping at budget: no rebound
- ₹ 3 MSEDCL pays 50% of the saving; booked at ₹37,574

|             |                                                                |
|-------------|----------------------------------------------------------------|
| Gujarat SKY | Paid for power; 60% kept pumping                               |
| Punjab PBPK | ₹4/kWh saved; 4.69% enrolled                                   |
| RAYNEX      | Metered, automatic, default-on                                 |
| ⚠           | Bonus is only 3.4% of base benefit: payback 2.6 yrs without it |

→ Exhibit F1 Payoff Matrix, Slide 11

### Per Hectare Per Year, Against Baseline

| Metric                | Baseline    | RAYNEX      | Change           |
|-----------------------|-------------|-------------|------------------|
| Water Pumped          | 14,000 m³   | 9,100 m³    | −35%             |
| Free Grid Power       | 2,120 kWh   | 0, solar    | −2,120 kWh       |
| Tomato Water Use      | 3.57 kg/m³  | 6.59 kg/m³  | + 85%            |
| Yield, Tomato / Onion | 25 / 18 t   | 30 / 21.6 t | + 20%            |
| Loss, Tomato / Onion  | 11.6% / 10% | 7.7% / 6.7% | 1.88 t saved     |
| Energy Per Quintal    | 4.93 kWh    | 3.74 kWh    | −24%             |
| Grid Export           | 0           | 884 kWh     | + 884            |
| CO₂ Avoided           | n/a         | 2.14 t      | 2.14 t           |
| Net Benefit           | n/a         | ₹1,37,693   | ₹81,150 loan yrs |

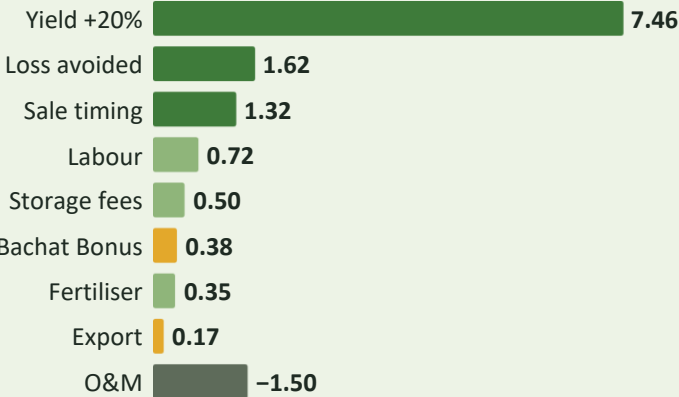
- 💧 Drip needs 10 m more head, so 35% less water saves 9% of pump energy; the sun supplies the rest
- ↘ Conservative (+10% yield, no bonus, no sale timing): ₹54,810 a hectare

→ Exhibit A1 And A4, Slide 12

### Where The Money Comes From

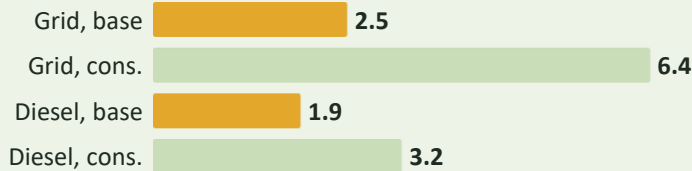
Exhibit A7, Slide 12

Per hub of 8 ha, ₹ lakh a year



|                       |                              |
|-----------------------|------------------------------|
| Net, base<br>₹11.02 L | Net, conservative<br>₹4.38 L |
|-----------------------|------------------------------|

Payback on ₹28.06 L net capex, years



Per Hub → At 500 Hubs

|                             |                        |                          |                  |                         |
|-----------------------------|------------------------|--------------------------|------------------|-------------------------|
| 💧 Water 39,200 → 1.96 cr m³ | 🌿 CO₂ 17.1 t → 8,529 t | 🍅 Food 15.04 t → 7,520 t | 👥 Jobs 1.2 → 600 | 🏠 MSEDCL saves ₹4.70 cr |
|-----------------------------|------------------------|--------------------------|------------------|-------------------------|

Every Player Gains  
a year, per hub

|                                        |                           |                          |                                    |                                     |                             |                             |
|----------------------------------------|---------------------------|--------------------------|------------------------------------|-------------------------------------|-----------------------------|-----------------------------|
| 👤 Farmer<br>+₹32,460, then<br>+₹55,077 | 👥 FPO<br>Fee book ₹6.02 L | 🏢 MSEDCL<br>Net +₹46,968 | 🏦 Bank<br>₹25.25 L, Gol<br>pays 3% | 🌞 Rural Youth<br>₹15,000 a<br>month | ⚡ Schneider<br>₹1.2 L a hub | 💧 Aquifer<br>39,200 m³ kept |
|----------------------------------------|---------------------------|--------------------------|------------------------------------|-------------------------------------|-----------------------------|-----------------------------|

One district with three farm zones proves the hub works on every soil; existing schemes and a farm loan pay for it; partners each bring one part.

3 hubs in 3 Nashik talukas from April 2027, ₹0 farmer capital, 500 hubs by year 5.

D6: Deployment And Scale-Up · E8: Roadmap

Why Nashik: Three Zones, One District

Exhibit A8, Slide 12

|                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                |                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>📍</div><div>Hub 1: Igatpuri</div><div>Western Ghat zone</div><div>Red, lateritic soil · very high rain rain</div><div>Groundwater safe</div><div>Paddy on AWD + vegetables</div><div>Tests: wet, hilly hub</div></div> | <div><div>📍</div><div>Hub 2: Niphad</div><div>Plains, irrigated</div><div>Medium-deep black soil</div><div>Groundwater 97.4%, critical</div><div>Onion, tomato, grape · Lasalgaon</div><div>Tests: anchor grid hub</div></div> | <div><div>📍</div><div>Hub 3: Yeola</div><div>Scarcity zone</div><div>Shallow soil · low rain</div><div>Groundwater 79.0%, semi-critical critical</div><div>Onion, rainfed crops</div><div>Tests: drought hub, farm pond</div></div> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

- 📖 Nashik spans 3 of Maharashtra's 9 agro-climatic zones and 550 to 3,400 mm of rain: a "mini-Maharashtra"
- 🕒 58.8 lakh t of onion a year and Lasalgaon, Asia's largest onion market; free power for 44.06 lakh state pumps

What It Costs And Who Pays

Exhibit A6, A7, Slide 12

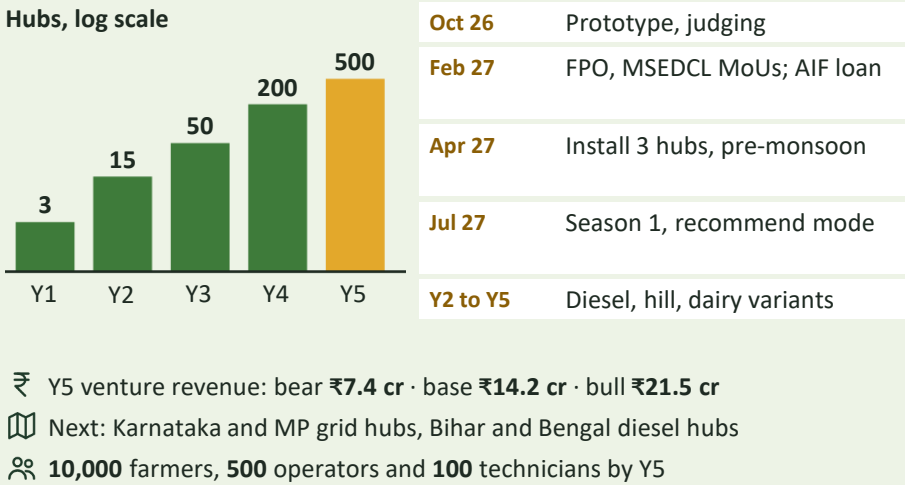
|                                                                                                       |                                  |
|-------------------------------------------------------------------------------------------------------|----------------------------------|
| Schemes ₹9.10 L                                                                                       | AIF loan ₹25.25 L at 6%, 7 years |
| Gross capex per hub ₹37.16 L; farmer capital ₹0                                                       |                                  |
| FPO grant ₹2.81 L                                                                                     |                                  |
| ☀️ PV: KUSUM 2.0, about ₹50,000 cr with 10 GW agri-PV, awaits notification; fallback Magel Tyala, 90% |                                  |
| 💧 Drip: Per Drop More Crop, 55% for small farmers                                                     |                                  |
| ❄️ Cold room: MIDH 35%; up to 75% stacked with RKVY (upside)                                          |                                  |
| 🏠 Loan: AIF, 3% interest subvention and a credit guarantee                                            |                                  |
| ₹ Repayment: fee ₹30,117 per farmer a year covers the ₹4.52 L instalment and O&M                      |                                  |
| Pilot of 3 hubs: ₹1.11 Cr gross, ₹84 L net of subsidy, all booked at 80%                              |                                  |

Partnerships, Each With A Fallback: No Grant Asked

| Partner                 | Brings                               | Gains                                | Fallback              |
|-------------------------|--------------------------------------|--------------------------------------|-----------------------|
| Schneider Electric      | ATV320, M262, meters, design support | A DISCOM product, a reference design | Ecozen Ecotron        |
| MSEDCL, Nashik Zone     | Net meter, ₹3.05 export, bonus MoU   | ₹46,968 net a hub                    | Season-1 bonus waived |
| Sahyadri Farms FPC      | 30,000 farmers, market access        | Storage, member value                | A Niphad FPO          |
| KVK Nashik, MPKV Rahuri | Agronomy, field trials               | Field data                           | NHRDF                 |
| NABARD, AIF Lender      | ₹25.25 L loan, guarantee             | Subvented, secured loan              | SFAC                  |
| Ecozen                  | Ecofrost cold room                   | Install base                         | Inficold              |
| Sarvam, Bhashini        | Voice in 22 languages                | An agri use case                     | AI4Bharat models      |
| Lasalgaon APMC, eNAM    | Prices, buyers                       | Steadier arrivals                    | FPO pooled sales      |

Rollout And Roadmap

Exhibit F8, Slide 11

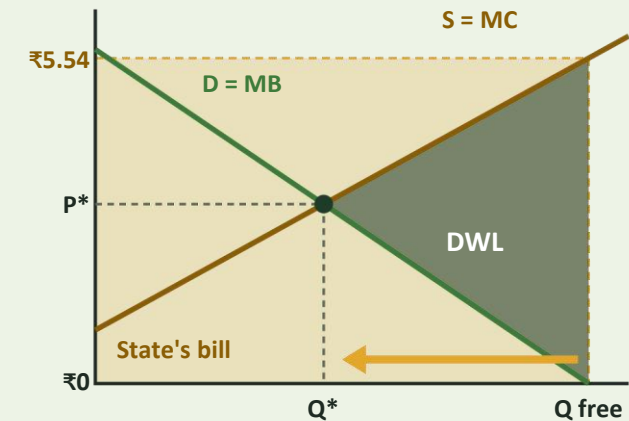


# EXHIBITS F1 TO F8: THE FRAMEWORKS BEHIND THE CASE

Public policy, micro and macroeconomics, strategy and projections: why the state should pay, how much, and why the numbers hold.

## F2 Microeconomics: The Subsidy Graph

Farm power: ₹/kWh vs kWh pumped



Gold arrow: the bonus pulls use back to Q\*

- ⚡ **Free power:** the farmer pays ₹0, so he pumps to Q free, where the last unit is worth nothing to him
- ₹ **State's bill:** MSEDCL still pays ₹5.54 a unit; the state covers the box, ₹14,761 cr a year
- ⚠ **Deadweight loss:** every unit past Q\* costs more than it is worth
- 🔧 **Our fix:** a Pigouvian subsidy on saving. Each unit not pumped earns ₹2.77, half of ₹5.54, so use moves back toward Q\*

📄 **Evidence:** Punjab's ₹4 a unit cut pumping 7.5 to 30%

## F3 Public Policy: How Much Should The State Pay?

| Government Ledger, Per Hub   | One-Time | Every Year            |
|------------------------------|----------|-----------------------|
| Capex: KUSUM 2.0, PDMC, MIDH | ₹9.10 L  | n/a                   |
| Power purchase avoided       | n/a      | + ₹93,936             |
| Bachat Bonus to FPO          | n/a      | − ₹46,968             |
| AIF 3% subvention, GoI       | n/a      | − ₹75,756 Y1, falling |
| Net to MSEDCL                | n/a      | + ₹46,968             |

🔧 **Our call:** no new capex money. Bonus = 50% of the metered saving, paid after the season

- 🎯 **Bounds:** cost of saving < bonus ≤ ₹93,936; 50% splits the surplus (Nash bargain)
- 💧 **Water price:** ₹0 → a shadow price of ₹1.20 per m<sup>3</sup> saved (₹46,968 ÷ 39,200 m<sup>3</sup>)
- 🍅 **Food prices:** stored lots sell +₹3/kg off-peak, easing Lasalgaon gluts
- ⚡ **Power price:** MSEDCL buys day solar at ₹3.05, below its ₹5.54 average

💡 **Self-financing:** the state pays for water not used out of power it no longer buys; states spent ₹1.89 lakh cr on power subsidy in FY25.

## F1 Game Theory: Payoff Matrix

FPO / MSEDCL payoff, ₹ per hub a year

|      | Bonus             | None         |
|------|-------------------|--------------|
| Save | +46,968 / +46,968 | 0 / +93,936  |
| Pump | 0 / 0             | 0 / 0, today |

🎯 Without a bonus, pumping is dominant; with one, (save, pay) is the Nash equilibrium

✅ MSEDCL pays only on proof

**Who moves first:** FPO grant and AIF fund 3 pilots → meters prove → MSEDCL signs → AIF lends hubs 4 to 50

🔄 **Repeated game:** a payout each season builds trust

## F8 Projections And Viability

Market, ₹ crore, log scale

|     |        |
|-----|--------|
| TAM | 19,600 |
| SAM | 18,000 |
| SOM | 185.8  |

TAM: 49 lakh KUSUM pumps · SAM: 45 lakh Maharashtra farm pumps · × ₹40,000

**Hubs Y1 to Y5: 3 · 15 · 50 · 200 · 500**  
**Farmers: 60 → 10,000**

**Y5 revenue:** bear ₹7.4 cr · base ₹14.2 cr · bull ₹21.5 cr

🔧 **MSEDCL at 500 hubs:** avoids ₹4.70 cr, pays ₹2.35 cr a year

🕒 **Viability:** ₹11.02 L a year on ₹28.06 L net: payback 2.5 yrs

## F4 Lessons From Abroad And Home

|         |                                                        |
|---------|--------------------------------------------------------|
| Denmark | 1,163 co-op dairies by 1909: copy one member, one vote |
| Israel  | Quotas, block rates: copy the budget                   |
| Gujarat | Paid for power only: avoid                             |
| Punjab  | 4.69% enrolled, late pay: make it default, automatic   |
| China   | 2.51 lakh drones on 178 mn ha: share machines          |

## F6 JTBD And Rogers Diffusion

- 🔧 **JTBD:** sleep, safety, a job, a smaller bill
- ➡ **Rogers:** FPO board first; recommend mode to try; neighbour plots to see

## F5 F7 Strategy, Macro, PESTEL

**Now:** KUSUM 2.0 pending

**1 to 5 yrs:** ₹50,000 cr, 10 GW agri-PV

**5 to 15 yrs:** free power ends 2029

**P** Free farm power      **E** ₹1.89 lakh cr

**S** 82% smartphones      **T** 22-language AI

**E** Aquifers falling      **L** DPDP, ALMM-II

🎯 **7 Powers:** counter-position, switching costs, feeder data

🔧 **Porter:** rivals Ecozen, Fasal; barrier, the DISCOM contract

⭐ **Blue Ocean:** create the bonus, eliminate night pumping

✅ **Moat:** the DISCOM contract, not the box

# EXHIBITS A1 TO A10: EVERY ASSUMPTION, IN ONE PLACE

Every number in slides 2 to 10 traces to a line here; (A) marks our model, the rest are published figures. Scheme money is booked at 80%.

## A1 Baseline, Per Hectare

|                       |                   |
|-----------------------|-------------------|
| Crops                 | Tomato K, onion R |
| Flood Water, A Season | 7,000 m³          |
| Water A Year          | 14,000 m³         |
| Pump Lift (A)         | 25 m              |
| Grid Power A Year     | 2,120 kWh         |
| Farm-Gate Cooling     | None              |
| Yield, Tomato / Onion | 25 / 18 t         |
| Energy Per Quintal    | 4.93 kWh          |

## A5 Loss And Storage

|                          |               |
|--------------------------|---------------|
| Loss, Tomato / Onion (A) | 11.6% / 10%   |
| Loss Cut (A)             | One-third     |
| Food Saved Per Hub       | 15.04 t       |
| Sale Timing, Stored      | ₹3/kg on 44 t |
| Storage Fees             | ₹50,000       |
| Crates Guarded           | 250 × 20 kg   |
| Cold Room Draw           | 15 kWh a day  |
| Onion Loss, NABCONS      | 7.27%         |

## A9 Manpower, O&M, Drones

|                        |                 |
|------------------------|-----------------|
| Operator, Per Hub      | ₹4,000 a month  |
| Technician, Per 5 Hubs | ₹15,000 a month |
| Platform Fee           | ₹60,000 a year  |
| Spray Drone, Certified | ~₹10 L          |
| O&M Total, Per Hub     | ₹1.50 L a year  |
| Drone Didi Subsidy     | 80%, ₹8 L cap   |
| Team Y1, Y3, Y5        | 5 / 12 / 35     |
| Labour Cost Cut, Drip  | 30 to 40%       |

## A2 Farm And Hub

|                     |                 |
|---------------------|-----------------|
| Farmers Per Hub     | 20              |
| Area Per Hub        | 8 ha            |
| Ramesh's Plot       | 0.4 ha          |
| Ramesh's Revenue    | ₹1,86,400       |
| Hub Crop Revenue    | ₹37.28 L        |
| Tomato Price        | ₹10/kg          |
| Power To The Farmer | Free to 7.5 HP  |
| FPO Equity Grant    | ₹2,000 a member |

## A6 Policy Money, Booked At 80%

|                          |                |
|--------------------------|----------------|
| KUSUM 2.0 Or Magel, PV   | 30%            |
| PDMC, Drip               | 55%            |
| MIDH, Cold Room          | 35%            |
| Booked Subsidy           | ₹9.10 L        |
| MSEDCL Power Cost        | ₹5.54/kWh      |
| Bachat Bonus             | 50% of ₹93,936 |
| Bonus Booked             | ₹37,574        |
| Equity Grant Cap Per FPO | ₹15 L          |

## A10 Rollout And Revenue

|                       |                         |
|-----------------------|-------------------------|
| Hubs Y1 To Y5         | 3, 15, 50, 200, 500     |
| Integration Margin    | ₹3.7 L a hub            |
| Schneider Hardware    | ₹1.2 L a hub            |
| Voice, Per 3-Min Call | ₹7 to 8                 |
| Install Window        | Apr to Jun              |
| Pilot Talukas         | Igatpuri, Niphad, Yeola |
| Margin Money, 10%     | ₹2.81 L                 |
| Y5 Revenue, Base      | ₹14.2 cr                |

## A3 Energy Model (A)

|                        |                    |
|------------------------|--------------------|
| Array                  | 15 kWp             |
| Yield Per kWp          | 1,700 kWh          |
| Generation A Year      | 25,500 kWh         |
| Water, Cold, Dryer kWh | 15,430, 2,700, 300 |
| Export At ₹3.05, MSKVY | 7,070 kWh          |
| Pump Efficiency        | 45%                |
| Grid CO₂               | 0.71 kg/kWh        |
| On-Site Use Vs Dhundi  | 72% vs 35%         |

## A7 Capex And Finance

|                         |                     |
|-------------------------|---------------------|
| PV, Pump, Drive         | ₹9.50 L             |
| Drip ₹1.12 L/ha × 8     | ₹8.96 L             |
| Cold Room, 5 t          | ₹12.00 L            |
| Borewell, Tank; Sensors | ₹2.5 L; ₹1.4 L      |
| Dryer, Crates; Scale    | ₹2.5 L; ₹0.3 L      |
| Gross / Net Capex       | ₹37.16 L / ₹28.06 L |
| AIF 6% For 7 Yrs: Loan  | ₹25.25 L            |
| Instalment; Per Farmer  | ₹4.52 L; ₹30,117    |

## A4 Water, Yield, Inputs

|                           |                |
|---------------------------|----------------|
| Water Saved, Base         | 35%            |
| Yield Gain, Base          | +20%           |
| Yield Gain, Conservative  | +10%           |
| Drip Extra Head (A)       | 10 m           |
| Fertiliser Saved          | 11% of ₹20k/ha |
| Labour Saved              | 30% of ₹15k/ha |
| Bulk Buying, Not Booked   | 10%            |
| Pump Energy Saved By Drip | 9%             |

## A8 Variants

|                         |                  |
|-------------------------|------------------|
| Diesel Saved, V2 Hub    | ₹4,53,600        |
| V2 Payback              | 1.9 yrs          |
| AWD Water / Methane Cut | 30% / 48%        |
| Membrane, V3            | ₹3.0 to 3.9 L/ha |
| Exchange Rate           | ₹87 per \$       |
| Hill Subsidy, V4        | 50%              |
| Agrivoltaic Yield Floor | 80%              |
| Diesel Hub CO₂ Saved    | 13.5 t           |

## A11 Sensitivity: Payback In Years, Yield Gain × Cold-Room Cost

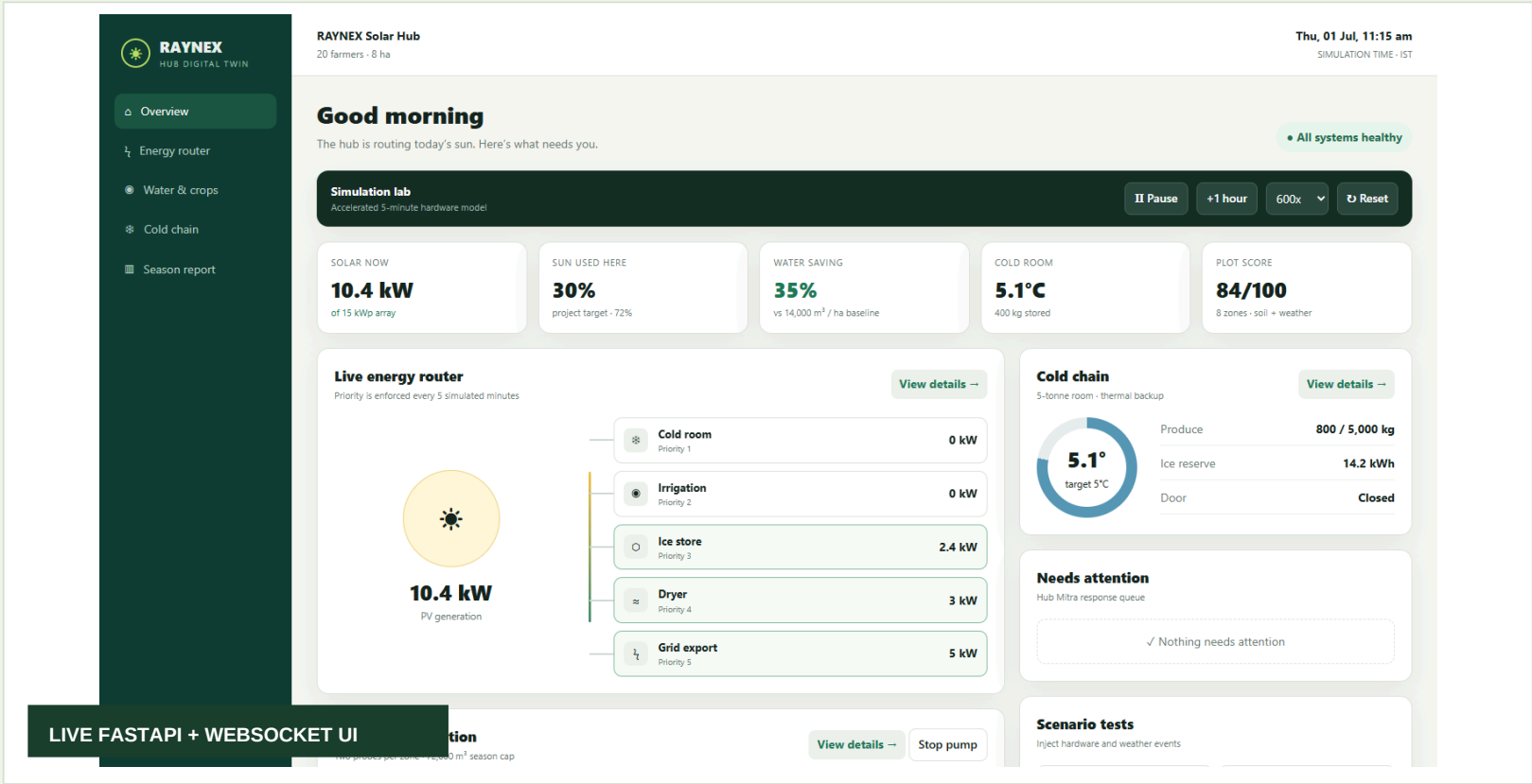
| Yield | ₹8 L | ₹12 L | ₹15.1 L | ✓ Every cell pays back inside the 7-year AIF loan                            |
|-------|------|-------|---------|------------------------------------------------------------------------------|
| 10%   | 3.5  | 3.9   | 4.2     | ⚠ Weakest case, 10% yield and a ₹15.1 L room: 4.2 years                      |
| 20%   | 2.3  | 2.5   | 2.7     | 🕒 No PV subsidy: 2.5 → 2.7 yrs; no bonus: 2.6 yrs                            |
| 30%   | 1.7  | 1.9   | 2.1     | ₹ Conservative stack: 6.4 yrs grid, 3.2 diesel; farmer at break-even (–₹693) |

# Simulation Lab: Prove The Router Before The Hardware

A five-minute digital twin turns the architecture into a testable operating system.

One screen validates energy priority, water logic, cold safety and farmer decisions.

D4: Prototype · E5: Technical Approach



## WHAT IT MODELS

- 15 kWp array - weather-shaped PV output
- Five jobs - cold > water > ice > dryer > grid
- Physical state - 8 zones, probes, pump, 5 t room

## HOW TO TEST

- Time - 1x, 60x, 600x or 3000x
- Faults - cloud, heat, outage, probe, open door
- People - approve, skip or stop the pump

## WHAT IT PROVES

- Priority - critical cold load always comes first
- Safety - daylight water + ice-backed cooling
- Evidence - exact kWh ledger and run report

D4 SOFTWARE  
PROTOTYPE  
  
Before hardware

1 INJECT

Change weather or break a sensor

2 OBSERVE

Watch power, soil and cold state

3 DECIDE

Approve or override the action

4 VERIFY

Export evidence against targets